Francis Alford

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Education

Columbia University — New York, NY
M.S. in Biostatistics | Expected May 2028 — *Aug. 2026 – Present*

Washington University in St. Louis — St. Louis, MO
B.S. in Computer Science and Mathematics, Minor in Philosophy — *Aug. 2022 – May 2026*

Research Experience & Projects

Research Assistant — Dec. 2025 – June 2026
Washington University in St. Louis, Prevention Research Center — *St. Louis, MO*

- Cleaned and analyzed demographic and public health data for 36 federally funded Choice Neighborhoods Implementation sites across 24 states
- Developed a Python-based interactive research toolkit to visualize and compare neighborhood demographic and health indicators against metropolitan-area benchmarks
- Collaborated with public health researchers to develop accessible data visualizations and comparison tools for researchers, community organizations, and housing stakeholders

Bike Share Feasibility and Spatial Analysis | *Python, QGIS, GeoPandas* — Summer 2025

- Developed predictive models to estimate bike-share demand using demographic and transportation data from multiple cities
- Analyzed geographic and socioeconomic patterns to identify areas for equitable bike-share expansion in St. Louis

NFL Yardage Distribution Prediction | *Python, PyTorch, LightGBM* — Spring 2026

- Conducted a faculty-supervised machine learning independent study using NFL player-tracking data to predict probability distributions of rushing yards
- Developed statistical baselines, multinomial logistic regression, LightGBM, and neural network models; engineered spatial features and evaluated performance using grouped cross-validation and Continuous Ranked Probability Score

Machine Learning Research, YODA Lab | *Python, PyTorch, Gymnasium* — Summer 2024

- Implemented Deep Q-Network and behavior-cloning models in PyTorch to study reinforcement and imitation learning
- Designed training and evaluation experiments in Gymnasium and compared model performance across experimental conditions

Additional Experience

Data Analyst Intern — March 2025 – May 2025
Being Patient — *Remote*

- Analyzed search and website performance data for a digital health publication using Google Search Console, SQL, and WordPress tools
- Implemented structured metadata and FAQ schema for health-related content to improve information accessibility and search visibility

Teaching Assistant — Aug. 2023 – Jan. 2025
Washington University in St. Louis — *St. Louis, MO*

- Supported approximately 200 students across three semesters of introductory computer science coursework in Java
- Led debugging and problem-solving support during labs and office hours and evaluated programming assignments for correctness and program design

Technical Skills

Languages: Python, R, SQL, Java, C++
Machine Learning: PyTorch, scikit-learn, LightGBM
Data Analysis: pandas, NumPy, Matplotlib
Geospatial Analysis: QGIS, GeoPandas
Tools: Git, Jupyter, Linux, Gymnasium